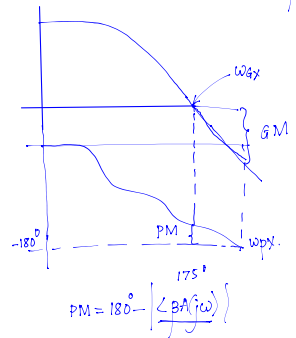


$$A_f(jw) = \frac{A(jw)}{1 + \beta A(jw)}$$



Phase margin = 5°

$\angle BA(jw) = -175^\circ$

$$A_f(jw) = \frac{A(jw)}{1 + \beta A(jw)}$$

$$= \frac{|A(jw)| e^{-j\angle A(jw)}}{1 + |\beta A(jw)| e^{-j\angle \beta A(jw)}}$$

$20 \log A(jw) - 20 \log \beta$

$20 \log \left(\frac{1}{\beta} \right)$

$A(jw)$

At $w = w_{gx}$

$|\beta A(jw)|_{w=w_{gx}} = 1$

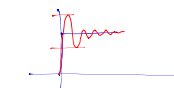
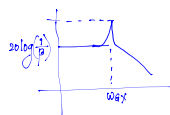
$|A(jw)|_{w=w_{gx}} = \frac{1}{\beta}$

$$|A_f(jw)|_{w=w_{gx}} = \frac{\frac{1}{\beta} |e^{-j175^\circ}|}{1 + |e^{-j175^\circ}|}$$

$$= \frac{\frac{1}{\beta} (\cos 175^\circ - j \sin 175^\circ)}{1 + |\cos 175^\circ - j \sin 175^\circ|}$$

$$= \frac{\frac{1}{\beta} (-0.0088 - j 0.087)}{1 + 0.088}$$

$$= \frac{17.5}{\beta}$$



Phase margin = 45°

$\angle = -135^\circ$

$|A(jw)|_{w=w_{gx}} = \frac{1}{\beta} |e^{-j135^\circ}|$

$= \frac{1}{\beta}$

Phase margin = 60°

$|A(jw)|_{w=w_{gx}} = \frac{1}{\beta}$

